

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Fiberglass Batts, target R-20.4 Roof upgrade: • Roof insulation: Blown Fiberglass, target R-34.5

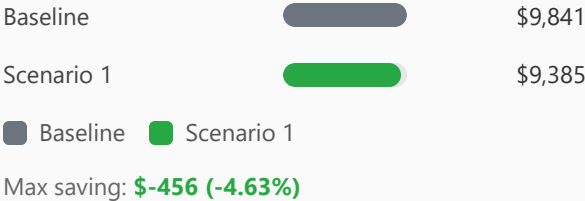
		Window upgrade: • 3-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: acrylic • Window frame replacement: wood window frame Door upgrade: • Door replacement: wooden door • Bottom seal: automatic door bottom • Top/side seal: jamb weatherstrip
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Annual Energy Analysis

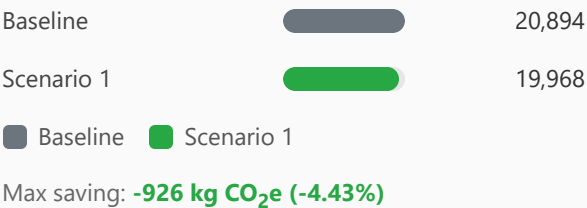
Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	222.59	13.95	5.90%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

\$3,226,579

Scenario 1

Min Retrofit Embodied Carbon

0.00 kg CO₂e

Embodied carbon intensity (ECI): 0.00 kgCO₂e/m²

Lowest Retrofit Cost Payback Period

7,081 years

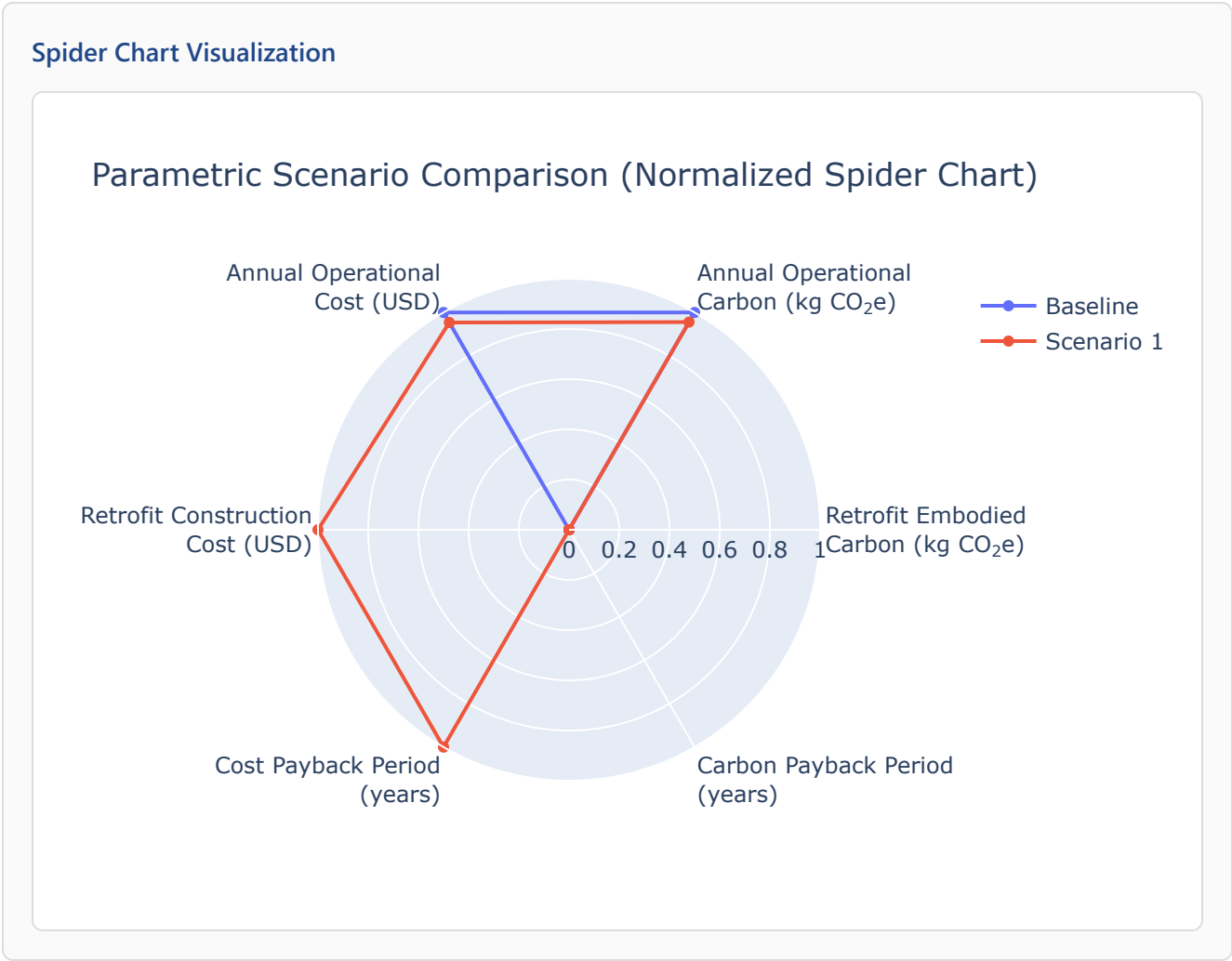
Scenario 1

Lowest Retrofit Carbon Payback Period

N/A

No valid carbon payback scenario

Comparative Visualizations between Renovation Scenarios



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 17,081 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1N/A

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	4.86	282	N/A	N/A	0.03	30	30
Scenario 1	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	30
Scenario 1	Window frame	wood window frame	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	wooden door	N/A	3.90	N/A	N/A	0.03	477	30
Scenario 1	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	jamb weatherstripping	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

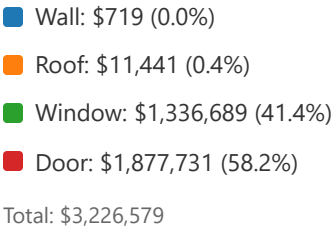
Scenario 1

Cost breakdown by retrofit measure



Carbon breakdown by retrofit measure

No retrofit measure contribution data.



Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	20,894	\$9,841
Scenario 1	0.00	\$3,226,579	19,968	\$9,385